Homework Assignment 3

Math 311, Fall 2026

Rutgers University

Exercise 2.2.7

(a) $(-1)^n$ is frequently in the set $\{1\}$ since any even n gives $(-1)^n = 1$. But $((-1)^n)$ is not eventually in $\{1\}$, since every odd-indexed term equals -1.

(b) Eventually is stronger since eventually $\implies$ frequently but not the converse.

(c) We can say (a_n) converges to a if, for every $\varepsilon > 0$, the sequence is eventually in $V_\varepsilon(a)$.

(d) Yes, (x_n) is frequently in $(1.9, 2.1)$ because infinitely many terms are equal to 2 in the interval. However, it might not be eventually. For example, consider

$$x_n = \begin{cases} 2, & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases}$$

Exercise 2.3.2

(a) Let $\varepsilon > 0$. Choose N such that, for $n \geq N$,

$$|x_n - 2| < \frac{3\varepsilon}{2}.$$

Then, for $n \geq N$,

$$\left|\frac{2x_n - 1}{3} - 1\right| = \left|\frac{2x_n - 4}{3}\right| = \frac{2}{3}|x_n - 2| < \frac{2}{3} \cdot \frac{3\varepsilon}{2} = \varepsilon.$$

Therefore,

$$\frac{2x_n - 1}{3} \to 1.$$

(b) Let $\varepsilon > 0$. Because $x_n \to 2$, choose N such that for $n \geq N$,

$$|x_n - 2| < \min\{1, 2\varepsilon\}.$$

The condition $|x_n - 2| < 1$ implies $x_n > 1$, so $|x_n| > 1$. Thus, for $n \geq N$,

$$\left|\frac{1}{x_n} - \frac{1}{2}\right| = \left|\frac{2 - x_n}{2x_n}\right| = \frac{|x_n - 2|}{2|x_n|} < \frac{|x_n - 2|}{2}.$$

Since $|x_n - 2| < 2\varepsilon$,

$$\left|\frac{1}{x_n} - \frac{1}{2}\right| < \frac{2\varepsilon}{2} = \varepsilon.$$

Hence,

$$\frac{1}{x_n} \to \frac{1}{2}.$$

Exercise 2.3.5 Let

$$(z_n) = (x_1, y_1, x_2, y_2, x_3, y_3, \ldots).$$

Suppose $(z_n) \to L$. Let $\varepsilon > 0$. There exists N such that if $m \geq N$, then

$$|z_m - L| < \varepsilon.$$

For $n \geq N$, we have $2n - 1 \geq N$, so

$$|x_n - L| = |z_{2n-1} - L| < \varepsilon.$$

Thus, $x_n \to L$. Similarly, if $n \geq N$, then $2n \geq N$, and hence

$$|y_n - L| = |z_{2n} - L| < \varepsilon.$$

So $y_n \to L$. Therefore, both (x_n) and (y_n) converge and have the same limit.

Conversely, suppose

$$x_n \to L \qquad \text{and} \qquad y_n \to L.$$

Let $\varepsilon > 0$. Choose N_1 such that if $n \geq N_1$, then $|x_n - L| < \varepsilon$, and choose N_2 such that if $n \geq N_2$, then $|y_n - L| < \varepsilon$. Let $N = 2 \cdot \max\{N_1, N_2\}$. Then for any $m \geq N$, we have even/odd cases.

If $m = 2k$, then $k \geq \max\{N_1, N_2\}$, so

$$|z_m - L| = |z_{2k} - L| = |y_k - L| < \varepsilon.$$

If $m = 2k - 1$, then $k \geq \max\{N_1, N_2\}$, so

$$|z_m - L| = |z_{2k-1} - L| = |x_k - L| < \varepsilon.$$

Hence $z_m \to L$.

Exercise 2.3.6 Notice

$$b_n = n - \sqrt{n^2 + 2n} = \frac{n^2 - (n^2 + 2n)}{n + \sqrt{n^2 + 2n}} = \frac{-2n}{n + \sqrt{n^2 + 2n}}.$$

Because $n > 0$, factor n out of the square root:

$$\sqrt{n^2 + 2n} = \sqrt{n^2\left(1 + \frac{2}{n}\right)} = n\sqrt{1 + \frac{2}{n}}.$$

Therefore,

$$b_n = \frac{-2n}{n + n\sqrt{1 + \frac{2}{n}}} = \frac{-2}{1 + \sqrt{1 + \frac{2}{n}}}.$$

Now,

$$\frac{1}{n} \to 0,$$

so by the Algebraic Limit Theorem,

$$1 + \frac{2}{n} \to 1.$$

and

$$\sqrt{1 + \frac{2}{n}} \to \sqrt{1} = 1.$$

Thus the denominator converges:

$$1 + \sqrt{1 + \frac{2}{n}} \to 1 + 1 = 2.$$

Using the quotient rule from the Algebraic Limit Theorem,

$$\lim_{n\to\infty} b_n = \frac{-2}{2} = -1.$$

Hence $\lim b_n$ exists and equals -1.

Exercise 2.3.8

(a) Since $x_n \to x$, the Algebraic Limit Theorem gives

$$x_n^k \to x^k \qquad (k = 1, 2, \ldots, d).$$

Hence

$$a_k x_n^k \to a_k x^k$$

for every k. Therefore,

$$p(x_n) = a_0 + a_1 x_n + \cdots + a_d x_n^d \to a_0 + a_1 x + \cdots + a_d x^d = p(x).$$

(b) Let

$$f(t) = \begin{cases} 0, & t = 0, \\ 1, & t \neq 0, \end{cases}$$

and choose

$$x_n = \frac{1}{n}.$$

Then $x_n \to 0$, but $x_n \neq 0$ for every n. Hence

$$f(x_n) = 1$$

for all n, so $(f(x_n)) \to 1$. However,

$$f(0) = 0.$$

Thus $f(x_n)$ converges, but not to $f(x)$.

Exercise 2.3.11

(a) Suppose $x_n \to x$, and let

$$y_n = \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

Then

$$y_n - x = \frac{(x_1 - x) + (x_2 - x) + \cdots + (x_n - x)}{n}.$$

Let $\varepsilon > 0$. Since $x_n \to x$, there exists N such that if $n \geq N$ then

$$|x_n - x| < \frac{\varepsilon}{2}.$$

Define

$$C = \sum_{k=1}^{N-1} |x_k - x|.$$

For $n \geq N$, we have

$$\begin{aligned} |y_n - x| &\leq \frac{1}{n} \sum_{k=1}^{n} |x_k - x| \\ &= \frac{C}{n} + \frac{1}{n} \sum_{k=N}^{n} |x_k - x|. \end{aligned}$$

Every term in the second sum is less than $\varepsilon/2$, so

$$\frac{1}{n} \sum_{k=N}^{n} |x_k - x| \leq \frac{n - N + 1}{n} \cdot \frac{\varepsilon}{2} \leq \frac{\varepsilon}{2}.$$

Choose n large enough that $C/n < \varepsilon/2$. Then

$$|y_n - x| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Therefore $y_n \to x$.

(b) Let

$$x_n = (-1)^n.$$

This sequence does not converge. But its averages do:

$$y_n = \frac{(-1)^1 + (-1)^2 + \cdots + (-1)^n}{n}.$$

If n is even, the numerator is 0, so $y_n = 0$. If n is odd, the numerator is -1, so

$$y_n = -\frac{1}{n}.$$

Thus $y_n \to 0$, even though (x_n) does not converge.

Exercise 2.4.1

Let

$$\alpha = 2 - \sqrt{3}.$$

Notice that α satisfies

$$\alpha = \frac{1}{4 - \alpha},$$

because $\alpha^2 - 4\alpha + 1 = 0$.

(a) First,

$$x_2 = \frac{1}{4 - x_1} = \frac{1}{4 - 3} = 1.$$

Also, $\alpha < 1 = x_2$. Suppose $x_n > \alpha$ and $x_n \leq 1$. Then

$$x_{n+1} - \alpha = \frac{1}{4 - x_n} - \frac{1}{4 - \alpha} = \frac{x_n - \alpha}{(4 - x_n)(4 - \alpha)} > 0.$$

So $x_{n+1} > \alpha$. Also, if $x_n \leq 1$, then

$$x_{n+1} = \frac{1}{4 - x_n} \leq \frac{1}{3} < 1.$$

Hence, by induction, $\alpha < x_n \leq 1$ for every $n \geq 2$. Now, for $n \geq 2$,

$$x_{n+1} - x_n = \frac{1}{4 - x_n} - x_n = \frac{x_n^2 - 4x_n + 1}{4 - x_n}.$$

Factor the numerator:

$$x_n^2 - 4x_n + 1 = (x_n - (2 - \sqrt{3}))(x_n - (2 + \sqrt{3})).$$

Because

$$2 - \sqrt{3} < x_n \leq 1 < 2 + \sqrt{3},$$

the numerator is negative, while $4 - x_n > 0$. Hence

$$x_{n+1} - x_n < 0.$$

Therefore (x_n) is decreasing and bounded below by $2 - \sqrt{3}$. By the Monotone Convergence Theorem, (x_n) converges.

(b) If $x_n \to L$, then removing the first term does not change the limit.

(c) Taking limits in

$$x_{n+1} = \frac{1}{4 - x_n},$$

gives

$$L = \frac{1}{4 - L}.$$

Thus

$$L(4 - L) = 1,$$

so

$$L^2 - 4L + 1 = 0.$$

The possible solutions are

$$L = 2 \pm \sqrt{3}.$$

But $x_n \leq 1$ for all $n \geq 2$, so its limit must satisfy $L \leq 1$. Therefore,

$$\lim_{n\to\infty} x_n = 2 - \sqrt{3}.$$

Exercise 2.4.2

(a) The problem is that the argument assumes the limit exists. In fact,

$$y_1 = 1, \qquad y_2 = 3 - 1 = 2, \qquad y_3 = 3 - 2 = 1,$$

so the sequence is

$$1, 2, 1, 2, 1, 2, \ldots$$

and does not converge.

(b) Only if we prove convergence is true first. Let

$$\beta = \frac{3 + \sqrt{5}}{2}.$$

Notice that

$$y_1 = 1 < \beta.$$

Assume $1 \leq y_n < \beta$. Then

$$y_{n+1} = 3 - \frac{1}{y_n} \geq 2 > 1.$$

Also, because $0 < y_n < \beta$,

$$\frac{1}{y_n} > \frac{1}{\beta},$$

and hence

$$y_{n+1} = 3 - \frac{1}{y_n} < 3 - \frac{1}{\beta} = \beta.$$

Thus $1 \leq y_n < \beta$ for every n.

Also,

$$y_{n+1} - y_n = 3 - \frac{1}{y_n} - y_n = \frac{-(y_n^2 - 3y_n + 1)}{y_n}.$$

Since

$$y_n^2 - 3y_n + 1 = \left(y_n - \frac{3 - \sqrt{5}}{2}\right)\left(y_n - \frac{3 + \sqrt{5}}{2}\right),$$

and $1 \leq y_n < \beta$, we have $y_{n+1} - y_n > 0$. Thus (y_n) is increasing and bounded above, so it converges.

Let

$$L = \lim_{n\to\infty} y_n.$$

Then

$$L = 3 - \frac{1}{L},$$

so

$$L^2 - 3L + 1 = 0.$$

Therefore,

$$L = \frac{3 \pm \sqrt{5}}{2}.$$

Because $y_n \geq 1$, we have

$$\lim_{n\to\infty} y_n = \frac{3 + \sqrt{5}}{2}.$$

Exercise 2.4.3

(a) Define

$$x_1 = \sqrt{2}, \qquad x_{n+1} = \sqrt{2 + x_n}.$$

Notice

$$x_1 = \sqrt{2} < 2.$$

If $0 < x_n < 2$, then

$$x_{n+1} = \sqrt{2 + x_n} < \sqrt{4} = 2.$$

Thus $x_n < 2$ for every n. Also, when $0 < x_n < 2$,

$$2 + x_n > x_n^2,$$

because

$$2 + x_n - x_n^2 = (2 - x_n)(x_n + 1) > 0.$$

Taking square roots gives

$$x_{n+1} = \sqrt{2 + x_n} > x_n.$$

Hence (x_n) is increasing and bounded above by 2, so it converges. Let $L = \lim x_n$. Then

$$L = \sqrt{2 + L}.$$

Thus

$$L^2 - L - 2 = 0,$$

so

$$L = 2 \quad \text{or} \quad L = -1.$$

Since $x_n \geq \sqrt{2}$ for every n, we have $L \geq \sqrt{2} > 0$.

(b) Define

$$y_1 = \sqrt{2}, \qquad y_{n+1} = \sqrt{2y_n}.$$

Again, $y_1 < 2$. If $0 < y_n < 2$, then

$$y_{n+1} = \sqrt{2y_n} < \sqrt{4} = 2.$$

So $y_n < 2$ for every n. Furthermore, if $0 < y_n < 2$, then

$$2y_n > y_n^2.$$

Taking square roots gives

$$y_{n+1} = \sqrt{2y_n} > y_n.$$

Therefore (y_n) is increasing and bounded above by 2, and hence it converges. Let $L = \lim y_n$. Taking limits in

$$y_{n+1} = \sqrt{2y_n}$$

gives

$$L = \sqrt{2L}.$$

Thus

$$L^2 = 2L,$$

so $L = 0$ or $L = 2$. Since $y_n \geq \sqrt{2} > 0$,

$$\lim y_n = 2.$$

Exercise 2.4.8

(a) Let

$$s_N = \sum_{n=1}^{N} \frac{1}{2^n} = \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^N}.$$

This is a geometric sum, so

$$s_N = \frac{\frac{1}{2}(1 - (1/2)^N)}{1 - \frac{1}{2}} = 1 - \frac{1}{2^N}.$$

Therefore,

$$\lim_{N \to \infty} s_N = 1.$$

So the series converges, and

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = 1.$$

(b) Let

$$\begin{aligned} s_N &= \sum_{n=1}^{N} \left(\frac{1}{n(n+1)} \right) \\ &= \sum_{n=1}^{N} \left(\frac{1}{n} - \frac{1}{n+1} \right) \\ &= \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \cdots + \left(\frac{1}{N} - \frac{1}{N+1} \right). \end{aligned}$$

Everything cancels except the first and last terms:

$$s_N = 1 - \frac{1}{N+1} = \frac{N}{N+1}.$$

Hence,

$$\lim_{N \to \infty} s_N = 1.$$

So

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1.$$

(c) Let

$$s_N = \sum_{n=1}^{N} \log \left(\frac{n+1}{n} \right) = \sum_{n=1}^{N} \left(\log(n+1) - \log n \right).$$

Then

$$s_N = (\log 2 - \log 1) + (\log 3 - \log 2) + \cdots + (\log(N+1) - \log N).$$

Thus,

$$s_N = \log(N+1) - \log 1 = \log(N+1).$$

Since

$$\log(N+1) \to \infty,$$

the partial sums do not converge to a real number. Therefore the series diverges.